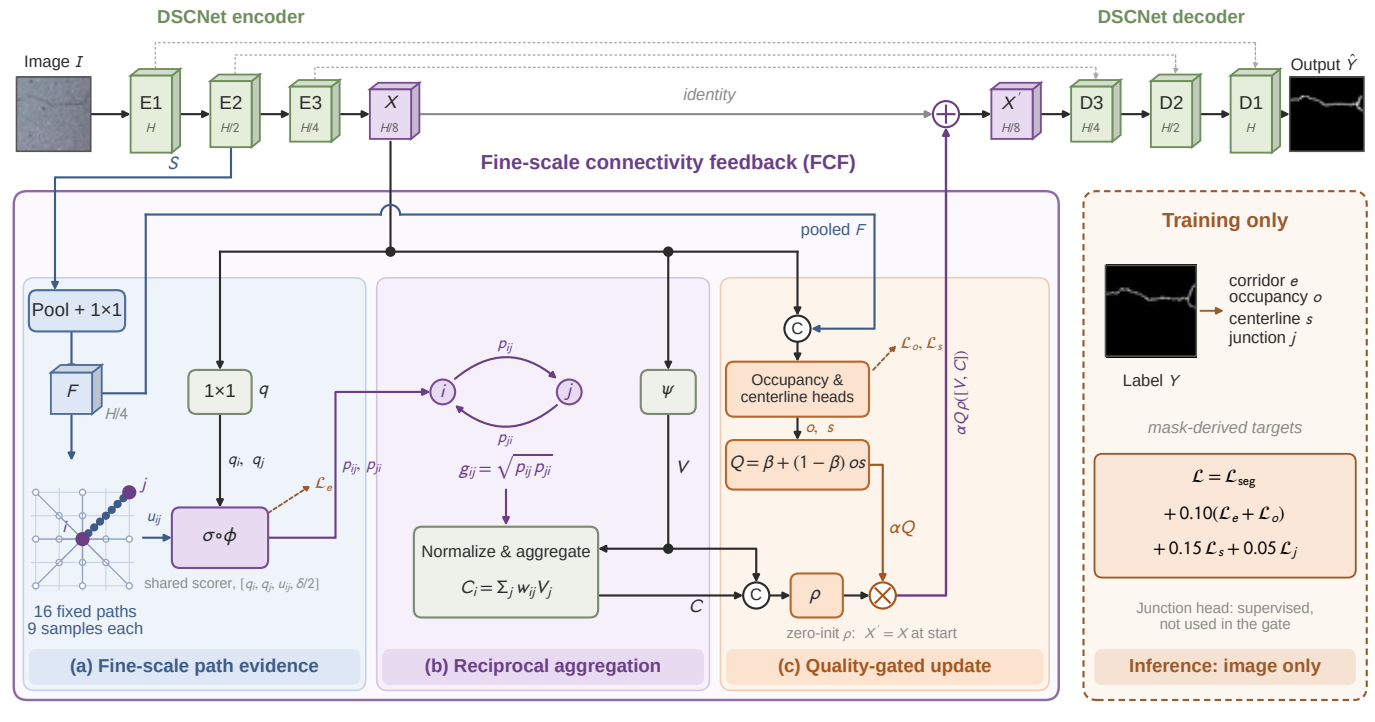

DSCNet encoder
DSCNet decoder
Image I
E1
H
E2
$H/2$
E3
$H/4$
X
$H/8$
S
identity
X'
$H/8$
D3
$H/4$
D2
$H/2$
D1
H
Output $\hat{Y}$
Fine-scale connectivity feedback (FCF)
pooled F
Pool + 1×1
F
$H/4$
1×1
q
q_i, q_j
u_{ij}
$\sigma \circ \phi$
$\mathcal{L}_e$
p_{ij}, p_{ji}
shared scorer, $[q_i, q_j, u_{ij}, \delta/2]$
16 fixed paths
9 samples each
(a) Fine-scale path evidence
p_{ij}
p_{ji}
$g_{ij} = \sqrt{p_{ij}\, p_{ji}}$
ψ
V
Normalize & aggregate
$C_i = \Sigma_j\, w_{ij} V_j$
C
(b) Reciprocal aggregation
C
Occupancy & centerline heads
$\mathcal{L}_o$, $\mathcal{L}_s$
o, s
$Q = \beta + (1-\beta)\, os$
αQ
ρ
$\alpha Q \rho(|V, C|)$
zero-init ρ: $X' = X$ at start
(c) Quality-gated update
Training only
corridor e
occupancy o
centerline s
junction j
Label Y
mask-derived targets
$\mathcal{L} = \mathcal{L}_{\text{seg}}$
$+ 0.10(\mathcal{L}_e + \mathcal{L}_o)$
$+ 0.15\, \mathcal{L}_s + 0.05\, \mathcal{L}_j$
Junction head: supervised, not used in the gate
Inference: image only